



Workshop on the Use of Alternative Data Sources for Consumer Price Statistics

UN Regional Hub for Big Data and
Data Science in Brazil

UN Task Team on Scanner Data

22–26 June 2026

 **UN Big Data Learning**

1. Overview of alternative data sources for CPI compilation

UN-CEBD Task Team
on Scanner Data

Traditional ways of collecting price data



To collect price information, the NSO would traditionally:

1. **Telephone** the shop
2. Search the **internet**
3. Issue **questionnaires**
4. Send **price collectors** to stores

Challenges with traditional methods:

- Time consuming
- Prone to manual/human errors
- Expensive
- Places limits on price collection



Alternative ways of collecting price data



Various alternative data sources (ADS) are currently being used by NSOs.

Challenges:

- Integration with existing methods
- Learning more advanced approaches to indices compilation
- Developing new skills

Compared to traditional methods:

- Less time-consuming if compared to manual human work of data collection
- Less error prone
- More affordable



Alternative ways of collecting price data



Various alternative data sources (ADS) are currently being used by NSOs.

Opportunities:

- In some case (scanner data) use of information on quantity sold (product-level turnover)
- Greater representativeness
- Improved timeliness
- Enabling NSO facing disruption in data collection (the case of COVID-19 crisis)

Benefits:

- Increased response rates
- Reduced respondent burden



What are the main characteristics of alternative data sources compared to the traditional one?

Data Dimension	Traditional	Scanner-data	Web-scraping	Administrative data
Data acquisition	Manual mainly via software running on tablet	Automated	Automated	Data acquired from government or other organizations collecting them mainly for non-statistical aims
Coverage	Sample of items Sample of retailers One/few monthly observations	All transactions Sample of retailer/ All 3 weeks or entire month	Bulk of products or sample Sample of online retailers	Census or almost census
Product identification	Item description Some attributes	Product ID, item description, some attributes Retailer specific product categories	Product ID, item description, more attributes Retailer specific product categories	Elementary units identified by key-codes and linked to products
Completeness of data	Only prices	Price and quantities / sales value and quantities sold	Prices and metadata No quantities	Prices, metadata and in some cases information on quantities sold
Timing	Single collection day	Daily/weekly	Daily/weekly	Weekly/monthly



Presentation contents

1. Scanner data characteristics
2. Web-scraped data characteristics
3. Characteristics of data obtained through APIs
4. Administrative data characteristics
5. Comparison among different ADS and features of an ADS project





 UN Big Data Learning

Scanner Data Characteristics

Contents

1. What are scanner data
2. NSOs' applications and use cases (Denmark and New Zealand)
3. Highlights



1. What are scanner data?

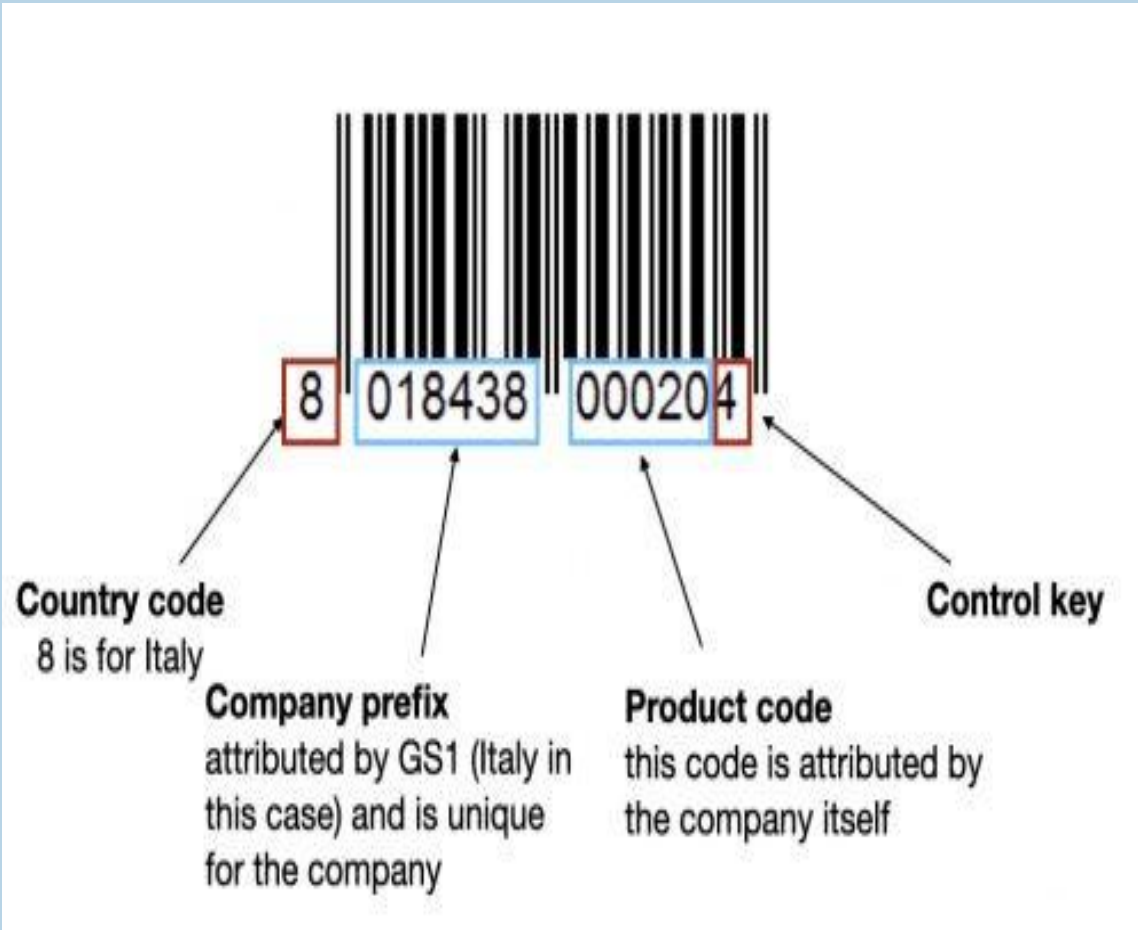


What are Scanner Data?

Scanner data are a kind of transaction data (records of actual purchases made by consumers) from purchases of goods in hypermarkets, supermarkets, petrol stations, and other outlets, where goods are sold via a scanned barcode.



Bar Codes



Any goods that display a bar code and which are scanned at the point of sale provide scanner data

Bar Code Standards:

- Set by an organization called “GS1”
- Each product assigned a Global Trade Item Number (GTIN)

Other codes:

- Stock-Keeping Unit (SKU)
- Universal Product Code (UPC)



Overview of Scanner Data characteristics

Here is an example of a scanner data set

Date	GTIN_number	Turnover	Units_sold	Unit_type	Base_size	Product_description	Shop
2011-04	2920080800007	3402.7	211	Gram	300	Bacon 2x150G	1
2011-04	2920080800007	2119.65	163	Gram	300	Bacon 2x150G	2
2011-04	2920080800007	1516.05	108	Gram	300	Bacon 2x150G	3
2011-04	2920080800007	1478.13	105	Gram	300	Bacon 2x150G	4
2011-04	2921056000005	302.5	14	Gram	200	Chicken filet	1
2011-04	2921056000005	102.5	5	Gram	200	Chicken filet	2



Benefits of transaction data

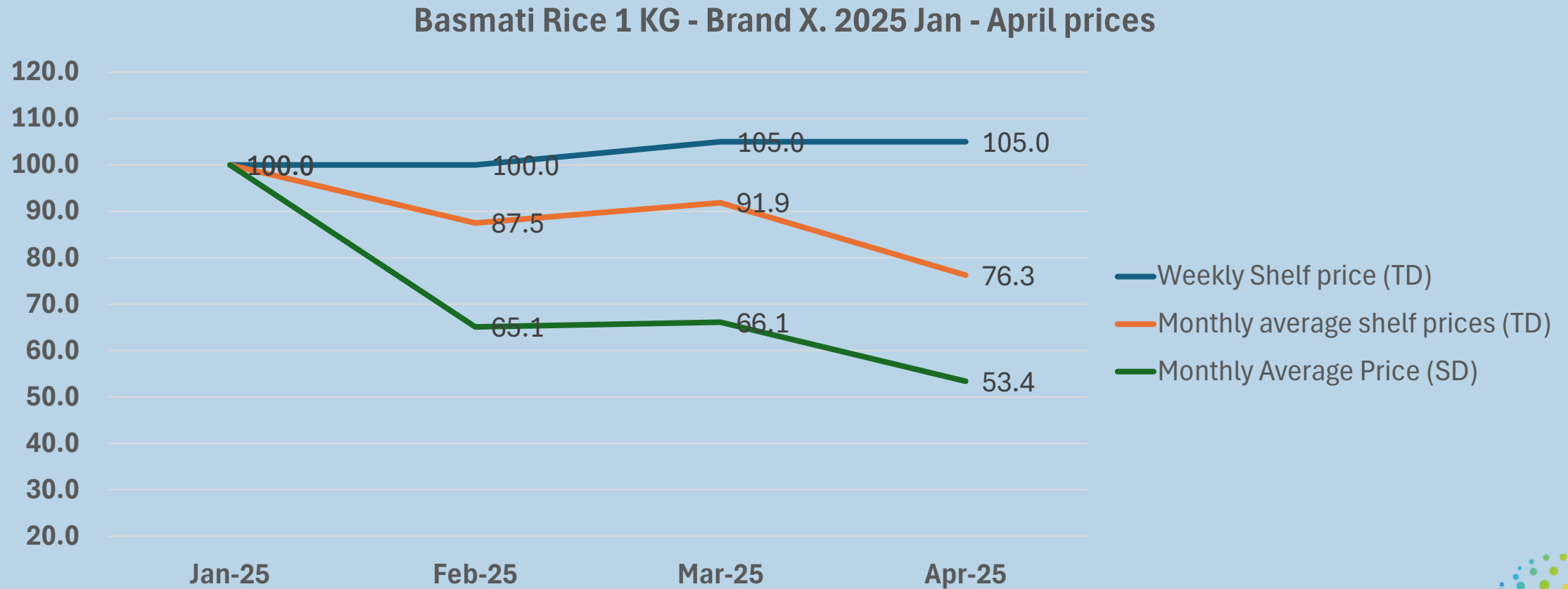
Why transaction data improves the quality of inflation measurement. An example (table)

1 Kg of Basmati Rice, brand X							
		Traditional Data collection (TD)		Scanner data			
		Weekly Shelf price (TD)	Monthly average shelf prices (TD)	Turnover	Quantity sold	Weekly Price (SD)	Monthly Average Price (SD)
25-Jan	Week1	100	100.0	1000	10	100	100.00
	Week 2	100		1200	12	100	
	Week 3	100		800	8	100	
	Week 4	100		400	4	100	
25-Feb	Week1	100	87.5	1000	10	100	65.12
	Week 2	100		1200	12	100	
	Week 3	50		3000	60	50	
	Week 4	100		400	4	100	
25-Mar	Week1	105	91.9	525	5	105	66.11
	Week 2	105		630	6	105	
	Week 3	105		315	3	105	
	Week 4	52.5		2100	40	52.5	
25-Apr	Week1	105	76.3	630	6	105	53.38
	Week 2	105		735	7	105	
	Week 3	55		2200	40	55	
	Week 4	40		2200	55	40	
	The week of monthly visit of data collector in case of once a month data collection						

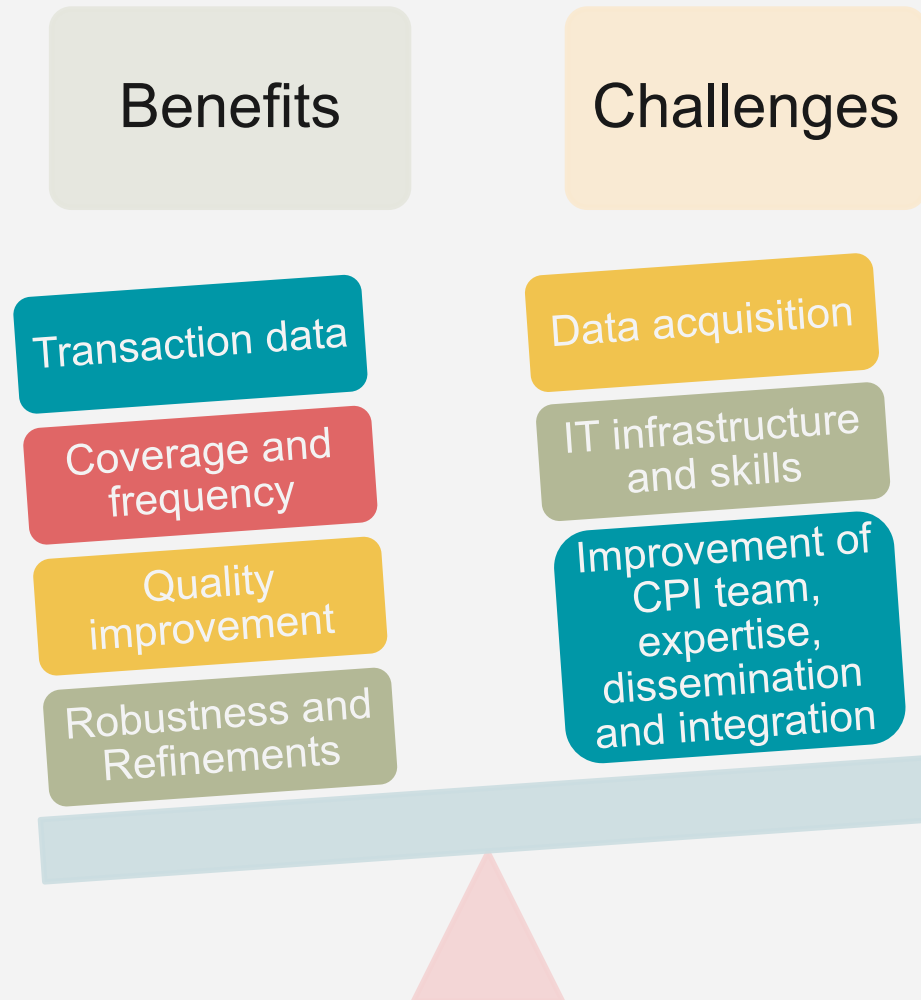


Benefits of transaction data

Why transaction data improves the quality of inflation measurement. An example (graph)



Benefits and challenges of Scanner Data





2. NSO applications and use cases



Is it just goods sold at hyper/supermarkets?



The number NSOs, using scanner data for CPI, continue to grow

NSOs tend to start collecting scanner data from hyper/supermarkets.

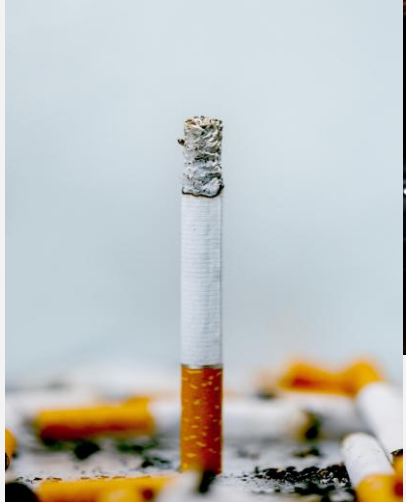
However, increasingly NSOs are seeking scanner data for less standard items from retailers such as:

- Petrol stations
- Pharmacies
- Electronic equipment stores

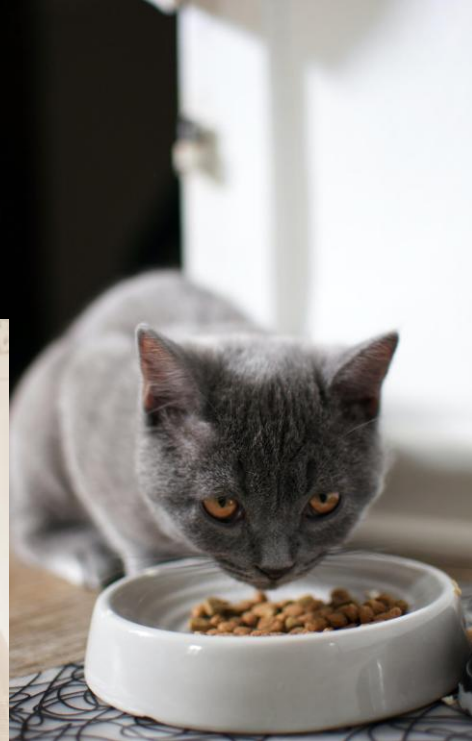


Statistics Denmark

- Scanner data has been received by Statistics Denmark from the main Danish supermarkets since 2011
- Since 2016, data are used for the production of indices for food, drinks and tobacco – Classification of Individual Consumption According to Purpose (COICOP) divisions 1 & 2
- Data are delivered directly via secure File Transfer Protocol (FTP) Servers



Statistics Denmark



- In 2020, Statistics Denmark increased the use of these scanner data to include non-food product (toilet paper, detergents, pet food, pregnancy tests and other products were included)





Statistics New Zealand

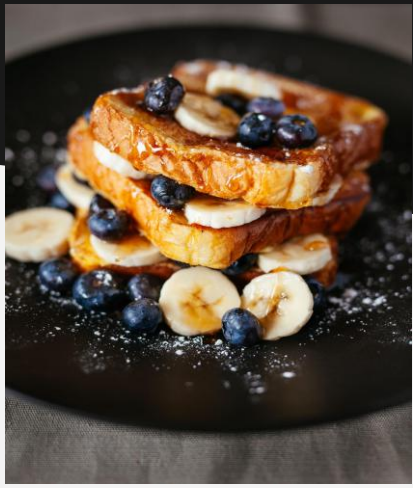


Consumer Electronic Scanner Data

- Since 2014 data were received from the Market Research company GfK (NZ is moving toward direct sourcing from retailers and diversified data inputs)
- Data provide a more complete picture of expenditure than traditional price collection methods
- Data include characteristics of each product, enabling explicit quality adjustment



Characteristics of scanner data received by NSOs



- Although data received by Statistics New Zealand and Statistics Denmark were obtained through different channels (directly for Statistics Denmark, indirectly via GfK for Statistics New Zealand) and used for different price index categories, in both the cases these data provide information on quantities sold and a more complete picture of expenditure (also in temporal terms) than traditional price collection methods.
- In some cases, they include information on the characteristics of each elementary items, enabling explicit quality adjustment, in other cases by using the bar code characteristics that can be captured online through automatic procedures (see the next slides on web scraping and APIs)





3. Highlights

Highlights

- Scanner data is a type of transaction data, generated in store or online
- Scanner data can provide more detailed data like turnover and quantity sold at the product level
- More sophisticated index calculations can be applied by the NSO
- Higher quality, more accurate price indexes can be produced
- Scanner data can increase response rates and reduce respondent burden
- Adopting scanner data may come with challenges.





UN Big Data Learning

**Web-scraped data
characteristics**

Contents

1. What web-scraped data is
2. NSOs' applications and use cases
3. Highlights



1. What web-scraped data is?



What web scraped data is

The screenshot shows a product page for a Lenovo notebook. The browser's developer tool is open at the bottom, displaying the DOM tree. The product name and price are circled in green on the page, and the corresponding JSON data in the developer tool is also circled in green.

Informatica > Computer Portatili > Notebook > LENOVO - NOTEBOOK IP 3 15ADA05 81W1018HIX-Platinum Grey

Consegna Gratuita

LENOVO - NOTEBOOK IP 3 15ADA05 81W1018HIX-Platinum Grey

Dimensione schermo-pollici: 15,6 pollici
Tipo di processore: Processore AMD Ryzen 7
Capacità RAM in GB: RAM 8 Gb
Capacità SSD-GB: SSD 512 GB

€ 697,00
IVA e contributo RAEE inclusi

Klarna. Paga in 3 rate da 232,33€ senza i
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Tufano Aprilia Srl

[Cambia negozio](#)

PRENOTA E RITIRA AGGIUNGI

Analisi pagina Console Debugger Rete Editor stili Prestazioni

```
<div class="col-lg-6 col-md-6 col-sm-6">  
  <div class="product-details-js" data-productinfo="{  
    "ean": "0195477721340", "name":  
    "LENOVO - NOTEBOOK IP 3 15AD...Notebook", "parentCategory": "Informatica", "quantity": "1"  
  }">  
  </div>  
</div>
```

- Data automatically collected from websites
- Focus on prices and e-commerce
- Information about products and services
- Potentially, unique product identifiers like bar codes
- Every website is different!



Information beyond prices

Groceries in an online supermarket



The screenshot displays a product page for De Cecco pasta. On the left is a vertical image of a blue and yellow bag of 'Penne Rigate n°41'. The bag features the 'DE CECCO' logo, 'dal 1886', a small illustration of a person, and a window showing the pasta. Text on the bag includes 'Penne Rigate n°41', '500g e', and 'MADE IN ITALY'. On the right, the product name 'De Cecco pasta penne rigate' is circled in green, followed by '500g' in a separate green circle. Below the name are five stars and '7 reviews'. The price '£2.00' is circled in green, with '£4/kg' next to it. A green 'Add to Trolley' button is next to a quantity selector set to '1'. Below this is a 'View all in Penne' button. The text 'Durum Wheat Semolina Pasta' is displayed. At the bottom, there are expandable sections for 'Nutrition' and 'Product Details', each with a downward arrow icon.

De Cecco pasta penne rigate 500g

★★★★★ 7 reviews

£2.00 £4/kg

1 Add to Trolley

View all in Penne

Durum Wheat Semolina Pasta

▼ Nutrition

▼ Product Details



Information beyond prices

Consumer Electronics e-shop

CUBOT X50 Smartphone Ohne Vertrag Android 11 Handy, 6,67 Zoll FHD+ Display, 8GB RAM + 256GB ROM, 4500mAh Akku, 32MP + 64MP Quad Kamera, 4G LTE Dual SIM, NFC, Face ID, GPS, grün

CUBOT
Simple & Trust

★★★★★ 28 Bewertungen



Bildschirmgröße

6,67 Zoll

Kameraauflösung

64 Megapixel

Betriebssystem

Android

Speicherkapazität

256 GB

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1 ▼

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Information beyond prices

Hotels

[Q Modifier la recherche](#)

Destination

Arrivée

Durée

Nombre de chambre(s)

Adulte(s)

Enfant(s)

Modifier la recherche >

Hammamet

06/08/2023

7

1 Chambre

2

0



La Playa Hotel Club ★ ★ ★

Petit déjeuner

Demi pension

Pension complete

All inclusive soft

Chambre1

☒ Chambre Double Standard

☐ Chambre Double Vue Mer

 Petit déjeuner

 Petit déjeuner

Disponible

Disponible

Moyen



651€

7 nuits en Petit déjeuner

+ De détails >



Advantages of web scraping data

Cost savings



More frequent data collection



Larger data set



Challenges and issues of web scraped data

Beyond advantages and benefits, there are also some drawbacks and challenges of web scraped data to consider:

- Web prices are not transaction data (no information on quantities sold)
- Caution should be taken when integrating web data into the CPI production (weights should be detected to reflect the importance of different data sources)
- Web scraping processes require specific IT skills, constant monitoring and maintenance and IT resources to store and process data.
- Often there is no clear regulation about web scraping operations, and the regulatory framework varies around the world.
- If a scraper is blocked, it is necessary to have a contingency plan to minimize the impact for the index production.



Possible applications for CPIs



- Automation of manual collection from websites
- Improvement of stores sample representativity
- Expansion of basket sectors
- Improvement of coverage of products in a given retailer
- Sample prices more frequently
- Extraction of rich metadata with product characteristics
- Targeted vs. Bulk web scraping



Benefits and challenges of web scraping

Benefits

Product attributes

Cost savings

Large data set

Frequency

Challenges

Regulation

IT skills and storing
resources needed

Not transaction
data



A stylized illustration on the left side of the slide. It features a shopping cart with a handle that is shaped like a globe. The globe has yellow and pink segments. The cart's body is light blue with a grid pattern. The wheels are pink. The background is a mix of pink, yellow, and light blue. A large green triangle is on the right side of the slide.

2. NSOs' applications and use cases



Web scraping of electricity and gas prices at Italian National Institute of Statistics (Istat)

Confronta le offerte

1

2

3

Tipo di offerta

Caratteristiche dell'offerta

Elenco delle offerte

CARATTERISTICHE DELLA RICERCA

Commodity: Energia elettrica Data: 12/07/2023

Tipo di offerta: Fissa Prezzo: Monoraria Livello di potenza: 3 kW

Consumo annuo: 2.700 kWh Tipo cliente: Domestico residente

CAP: 00185 - Roma

MODIFICA LA RICERCA

SONO STATE TROVATE **233** OFFERTE

Offerta variabile ↔ Ordina per... ▼

ULTERIORI FILTRI -

Nome offerta

Codice offerta

Categoria offerta

Tutte ▼

☐ Solo Flat

Sconti sempre applicabili

La spesa annua riportata rappresenta una stima e si basa su dati di consumo dichiarati dall'utente, o stimati, che si possono discostare dai consumi futuri e su prezzi che nel tempo possono variare.

Visualizza differenza di prezzo rispetto ai servizi di Tutela ☒

Esercente i servizi di tutela
Maggior Tutela

863,01 € annui

VAI AL DETTAGLIO →

- This is a case of targeted web scraping
- Gas and electricity retail prices on the free market are collected by Istat through a web site where all the operators are obliged to upload their offers allowing a transparent comparison
- Consumption parameters are setup for each city with specific profiles for yearly consumption, derived from other surveys
- Browser automation to collect data at regular intervals



Why airfares represent products for which web scraping is suitable

Collection of prices for the airfares is a suitable candidate for the use of web scraping. The main reasons (important parameters to evaluate the feasibility of the use of web scraping technique for a specific product group) are:

- The web is representative of the market given that consumers mostly buy airfares via the Internet
- A scraper can insert input parameters and extract easily the outputs related to a route
- This sector is where the competition is amongst a reduced number of airfares companies, and given that search engines are available to access airfares offered by different companies, this is also attractive as a reduced number of scrapers can be built and have good (or even full) coverage of the market



Web scraping of airfares at Brazilian Institute of Geography and Statistics (IBGE)



Inputs to define a flight:

Round trip ▾ 1 ▾ Economy ▾

Rio de Janeiro ↔ Buenos Aires EZE

Tue, Oct 17 < > | Tue, Oct 24 < >

[All filters](#) Stops ▾ Airlines ▾ Bags ▾ Price ▾ Times ▾ Emissions ▾ Connecting airports ▾ Duration ▾

Outputs: flights and prices according the input filters.

	6:55 AM – 9:50 AM Aerolineas Argentinas	2 hr 55 min AEP–GIG	Nonstop	149 kg CO ₂ Avg emissions ⓘ	R\$2,083 round trip	▾
	8:20 PM – 11:15 PM Aerolineas Argentinas	2 hr 55 min AEP–GIG	Nonstop	156 kg CO ₂ Avg emissions ⓘ	R\$2,083 round trip	▾



The background of the slide is a vibrant, abstract composition. On the left, a large, 3D-rendered folder icon in shades of pink, purple, and blue is the central focus. Behind it, there are blurred, colorful streaks and shapes in yellow, orange, and blue, suggesting a sense of motion or a digital environment. On the right side, a large, solid green triangle points towards the center. The text '3. Highlights' is positioned in the white space between the folder icon and the green triangle.

3. Highlights



Highlights

- Any website can be a source of data, bringing a rich set of product and service information
- Possibility for different kinds of applications and approaches (the choice amongst targeted and bulk extraction depends on the group of products to investigate, the website's characteristics, on the sampling strategy adopted and on the resources needed)
- Products representativity (no sales volume data)
- IT requirements (in term both of human and technical resources) have to be carefully analyzed in order to implement them
- Check for legal advice





**Application
Programming Interfaces
data characteristics**

Contents

1. What APIs data is
2. Benefits and challenges



1. What APIs data is



APIs data



- An API is a set of defined rules through which different applications communicate with each other.
- It can be seen as an intermediary that processes data transfers between systems
- Through APIs, the firms open their application data and functionality to external third-party developers, business partners, and internal departments
- In this module, we focus on APIs as a tool to access data on prices and characteristics of products and services

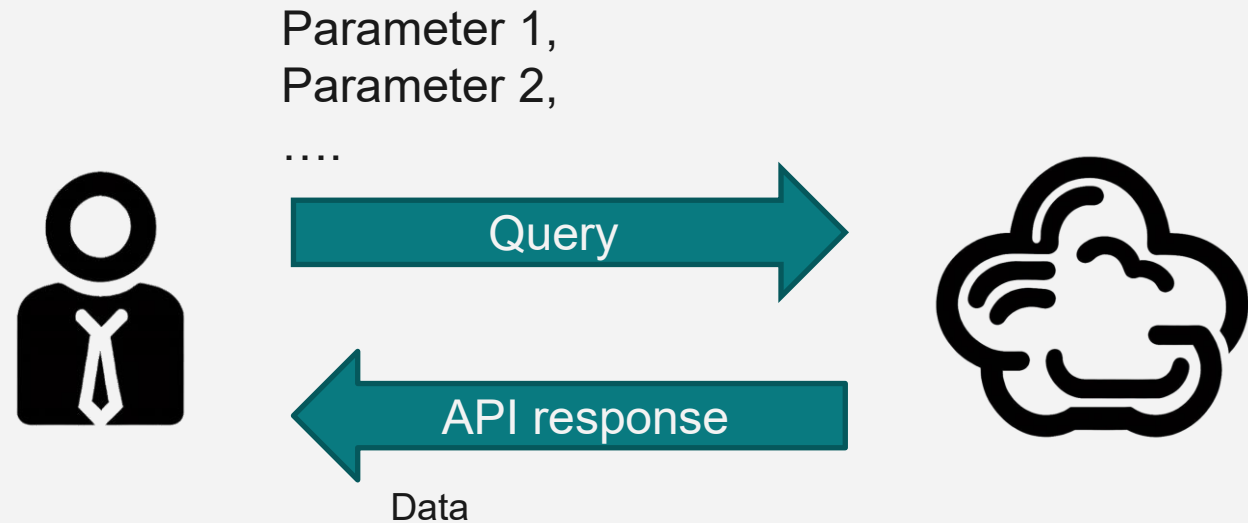


APIs models

Multiple pricing models



API calls



Advantages of API data

Cost and time savings
if compared with the traditional
price data collection



More frequent data collection

Stable data flow over time

More reliable and stable source
if compared with classical web-scraping



Possible applications for CPIs



- Automation of price data collection from websites
- Expansion of basket sectors
- Improve coverage of products in a given retailer
- Price data collection more frequent
- Extraction of rich metadata with product characteristics



Challenges and issues of API data

- API prices are not transaction prices but «shelf prices»
- Information on quantities sold usually are not available
- API providers can change pricing structure and charges over time with consequences on the NSOs' costs (APIs available for free could become available only on charge)
- Market coverage can be limited, considering that penetration of APIs could be quite different across products and services categories
- As in the case of web scraping, API processes also require specific IT skills, constant monitoring and maintenance and IT resources to store and process data



Benefits and challenges of API data

- APIs represent a suitable alternative to traditional web scraping to access price data on Internet allowing for an automated and stable data acquisition
- APIs suppliers set terms for data access, and this represents an advantage (the costs are predictable) but also disadvantages (terms and pricing may change over time in particular from free of charge to charge)
- Coverage varies by sector and market and API providers may not necessarily be representative of the market for a specific product or service category
- Skilled IT resources are needed at NSOs to access and process the data



Benefits and challenges of API data

Benefits

Product attributes

Frequency

Large data set

Stable access to web data

Challenges

IT skills and storing resources needed

No transaction data

Market coverage and pricing changes





Administrative Data Characteristics

Contents

1. What administrative data (advantages and disadvantages) is
2. NSO applications and use cases
3. Highlights



A large blue triangle pointing to the right, located on the left side of the slide.

1. What administrative is



What administrative is

- Administrative data are information collected by government departments and other organizations (public, private or third sector), primarily for operational reasons rather than statistical ones.
- In some cases, administrative data are collected for fiscal reasons (for example data referred to income or to rentals), in other cases for regulatory aims (for example insurance or fuels prices) or for registry reasons (such as the record of births and deaths). As such they are often census or almost census data



What administrative is

- Administrative data are commonly used by NSOs to build up statistical registers or to replace direct data collection in the area of business statistics
- For CPI purposes, administrative data are traditionally used to estimate the weights but in the recent years their use was extended also to the monthly calculation of price indices.
- This extension of their use for CPI was pushed by their increased digitalization and timeliness that made feasible their adoption for the monthly estimation of CPI



What administrative is

- While scanner and web-scraped data are primarily useful in the collection of data for goods (though can be useful for some service sectors too), administrative data are primarily useful in the collection of data for services (though can be useful for some goods sectors too).
- Unlike other data sources, the structure of administrative data is not prescribed and can vary depending on the original purpose they were collected for.



What are the main characteristics of administrative data compared to other alternative data sources ?

Data Dimension	Traditional	Scanner-data	Web-scraping	Administrative data
Data acquisition	Manual mainly via software ning on tablet	Automated	Automated	Data acquired from government or other organizations collecting them mainly for non- statistical aims
Coverage	Sample of items Sample of retailers One/few monthly observations	All transactions Sample of retailer/ All 3 weeks or entire month	Bulk of products or sample Sample of online retailers	Census or almost census
Product identification	Item description Some attributes	Product ID, item description, some attributes Retailer specific product categories	Product ID, item description, more attributes Retailer specific product categories	Elementary units identified by key-codes and linked to products
Completeness of data	Only prices	Price and quantities / sales value and quantities sold	Prices and metadata No quantities	Prices, metadata and in some cases information on quantities sold
Timing	Single collection day	Daily/weekly	Daily/weekly	Weekly/monthly



The main products or groups of products of CPI where administrative data are available and used



Housing (rental prices, dwelling stock)



Housing services (water, refuse collection, council tax)



Transportation services (rail, bus travel)



Communication services (internet, phone contracts)



Education (tuition fees)



Insurance (car, home, pet)



Energy (gas, electricity)



Fuel (petrol, diesel)



Motor vehicles (new cars, second-hand cars)



What are the main advantages of Administrative data for CPI aims?

- Improve data collection efficiency & cost-effectiveness. Administrative data are already collected even if for other aims
- They often bring metadata and information useful to their classification
- Reduction of respondent burden. Potential respondents are not requested another time to give the same information they have already provided
- Improve product and geographic coverage. Usually, administrative data cover the entire reference population
- Improve time coverage. Prices are referred to whole month/quarter
- Quantity information are sometimes available



What are the main challenges of Administrative data for CPI aims?

- Insufficient cooperation of the providers/holders of data could lead to poor quality
- Personal data protection issues could emerge and have to be faced
- Producers may not adhere to statistical standard concepts, definitions & classifications
- Data might already be processed and/or aggregated with the risk of unit value bias
- Sometimes data are not enough timely for the CPI monthly elaboration
- The data structure can vary depending on provider and original purpose of data, requiring a time-consuming preparatory work, including the setup of new methods or processing systems



Advantages and challenges of administrative data

Advantages

Better geographical and temporal coverage

Metadata and classification variables

Census/almost census data

Data already collected and reduction of respondent burden

Challenges

Issues of timeliness

No standard definitions and classifications

Scarce cooperation of data providers and personal data protection issues



2. NSO applications and use cases



The Italian case: rentals for housing

- The first use case related to an extensive use of administrative data that we focus on, is that of the acquisition of rentals for housing data base by the National statistical Institute of Italy (Istat)
- Istat receives the data set from the Tax Office that records this information mainly for taxation of the revenues deriving from the contract (a classic example of administrative data)



The Italian case: rentals for housing

- Since 2014 all the new contracts of rent of a real estate unit are registered through a unique online form and as such digitalized and they have to be registered with 1-month maximum delay
- Slowly the registration of the contracts has become more efficient and the cases of delays marginal
- Then, progressively, the availability of a long time series of data set of new contracts has allowed to detect, for the compilation of the monthly indices, not only the information on prices of the new contracts but also those of the existing ones



The Italian case: rentals for housing

- Therefore, the availability of good-quality data sets from 2017, and the research developed to elaborate correctly the information, has allowed Istat to implement the use of this source of data for CPI compilation, since January 2022
- In the following table are the main characteristics of the rentals data set obtained by Istat from Italian Tax Office



The Italian case: rentals for housing

Recap of the main characteristics of the database of rents received by Istat from Italian Tax Office

Data source	Database of new rental contracts (Tax Office)			
Data availability	From 2017			
Geographical coverage	Entire national territory (special case: Trentino)			
Frequency	Quarterly (monthly from the second quarter of 2022)			
Reference period	Quarter (month from the second quarter of 2022)			
Timeliness	35/40 days after the end of the reference quarter (it will reduced with the switch to the monthly provision of the data)			
Type of property	Residential (only whole apartment)	whole	A/1 – Luxury dwellings	
			A/2 – Well-finished (civil) dwellings	
			A/3 – Economic dwellings	
			A/4 – Popular dwellings	
			A/5 – Ultra-popular dwellings	
			A/6 – Rural dwellings	
			A/7 – Detached houses	
			A/8 – Villas	
			A/9 – Castles, palaces of great artistic and historical value	
			A/11 – Traditional local dwellings and accommodations	
	Non-residential (out of scope)			
Type of contract	Free contracts or agreed fee contracts; long or short term contract			
New contracts (thousands)	Year			
	2017	2018	2019	2020
	1,682	1,670	1,752	1,545
Target population (thousands)	1,417 (84%)	1,407 (84%)	1,486 (85%)	1,345 (87%)
Selected contracts (thousands)	813 (48%)	814 (48%)	844 (48%)	769 (50%)
Main variables	Dwelling location including OMI(1) zones; size in square meters; rent			

(1) An OMI segment is an aggregation of contiguous homogeneous zones. It represents a territorial area with specific geographical location in the municipality and reflects a consolidated urban location. Each municipal territory is divided into the following segments: Central, Semi-central, Peripheral, Suburban, Extra-urban



The Italian case: rentals for housing

- The micro data contain information of variables (in particular, the type of property, the size and the location of the dwelling) that are crucial to determine the rentals
- Therefore, Istat was able to identify homogeneous strata within which they could calculate for each quarter/month, the price per square meter to use to compile the indices



The Italian case: rentals for housing

- Both existing and new contracts contribute to the estimation of the price per square meters each month and the information about the dates of signature and the duration of the contracts make possible to manage correctly the contract that come in and those that go out

Since January 2022, Istat has been using the data provided by Italian Tax Office in the current estimation of monthly inflation



3. Highlights



Highlights

- Administrative data represent an emerging and important data source for CPI compilation that can replace the traditional data collection or better cover areas, where scanner data are not available or difficult to obtain and web scraping techniques inadequate
- In most of the cases they are census or almost census data (look at the cases of rentals for housing in Italy or that of rail fares in UK) and as such they guarantee an excellent coverage of the market, improving the quality of CPI when adopted



Highlights

- In some cases, they are transaction data containing information on expenditures and quantities purchased
- Considering that usually administrative data are data collected by others (for different aims), they can be accessed through one or a few data providers that are often government agencies or other bodies with a charge from government (facilitating the work of NSOs)
- Working with the government and the other data providers to agree the format and the variables to consider when the data are collected from the citizens (for instance introducing new variables in the collection for statistical purposes) can be crucial





**Comparison among
different ADS and
features of an ADS
project**

Contents

1. Comparative view of ADS according to different dimensions
2. Overview of main phases of an ADS project
3. Overview of core procedures in an ADS processing pipeline
4. Issues about data integration
5. Considerations for the design of an ADS project
6. Conclusions



Comparison among ADS

DIMENSION	TRADITIONAL	SCANNER	ADMINISTRATIVE	API	WEB SCRAPING
DATA ACQUISITION	MANUAL	AUTOMATED	AUTOMATED	AUTOMATED	AUTOMATED (ALTHOUGH REQUIRES MAINTENANCE FOR WEBSITE CHANGES)
COMPLETENESS/ SCOPE	SAMPLE ITEMS FROM SAMPLE OF RETAILERS	ALL TRANSACTIONS (CENSUS) FOR PARTICIPATING RETAILERS	SAMPLE OR UNIVERSE	BULK OR TARGETED SAMPLE FROM ONLINE RETAILERS	BULK OR TARGETED SAMPLE FROM ONLINE RETAILERS
METADATA	ITEM DESCRIPTION AND NSO SPECIFIED ATTRIBUTES	ITEM DESCRIPTION AND OFTEN LIMITED ATTRIBUTES	DEPENDS ON THE REGISTER	ITEM DESCRIPTION AND ATTRIBUTES AS DESCRIBED ON WEBSITE	ITEM DESCRIPTION AND ATTRIBUTES AS DESCRIBED ON WEBSITE
QUALITY	NONE (SHELF SPACE AND OTHER POSSIBLE PROXIES)	QUANTITIES SOLD	MAY CONTAIN QUANTITY	NONE (ALTHOUGH SEE SECTION DERIVING PROXY WEIGHTS FOR WEB SCRAPED DATA)	NONE (ALTHOUGH SEE SECTION DERIVING PROXY WEIGHTS FOR WEB SCRAPED DATA)
PRICE DATA	LIST PRICE	TRANSACTION PRICE	TRANSACTION PRICE	LIST PRICE	LIST PRICE
TIME OF DATA AVAILABILITY	BASED ON NSO PROCESSING SPEED	SCHEDULED FILE DELIVERIES	SCHEDULED FILE DELIVERIES	DEPENDENT ON AGREEMENT WITH SUPPLIER, BUT GENERALLY NEAR INSTANTANEOUS	NEAR INSTANTANEOUS
TIME COVERAGE	ONE POINT IN TIME IN COLLECTION PERIOD	COMPLETE, WITH THE POSSIBILITY OF HISTORICAL SERIES COVERING PERIOD BEFORE ACQUISITION	COMPLETE, WITH THE POSSIBILITY OF HISTORICAL SERIES COVERING PERIOD BEFORE ACQUISITION	DEPENDENT ON AGREEMENT WITH SUPPLIER, BUT GENERALLY HIGH FREQUENCY	HIGH-FREQUENCY (DEPENDENT ON THE PURPOSE, DIFFERENT FREQUENCIES ARE POSSIBLE EG SCRAPING CAN HAPPEN HOURLY, DAILY, WEEKLY OR MONTHLY)



Comparison among ADS

ADS RANK

Scanner/
Admin.
records

API

Web scraped

Power for CPI use

	Pros	CONS
Scanner/ administrative records	Transaction data, prices and quantities included , stores coverage/geographical breakdown, backward (in time) data, market coverage, stability of data transfer.	Data access needs to be negotiated , poor detail on product attributes.
API	Individual prices, Stability of data transfer. Rich in product attributes, high frequency of collection.	Data access needs to be negotiated, lack of quantities, market coverage, geographical breakdown
Web scraped	Individual prices, rich in product attributes, high frequency of collection, easy initial access.	Maintenance/instability of access, lack of quantities, market coverage, geographical breakdown



ADS phases

MAIN PHASES OF ADS PROJECT FOR CPI



Research



Implementation



Production





MAIN PHASES OF ADS PROJECT FOR CPI



Research

Design the project, elaborate and test methods for data processing and index calculation (including impact analysis), build the pieces of the new systems





MAIN PHASES OF ADS PROJECT FOR CPI



Implementation

Set the quality indicators for the chosen procedures, establish a fallback plan, evaluate the stability of the system and check if extra testing and needs are required. Check if staff need extra training for the routine changes.





MAIN PHASES OF ADS PROJECT FOR CPI

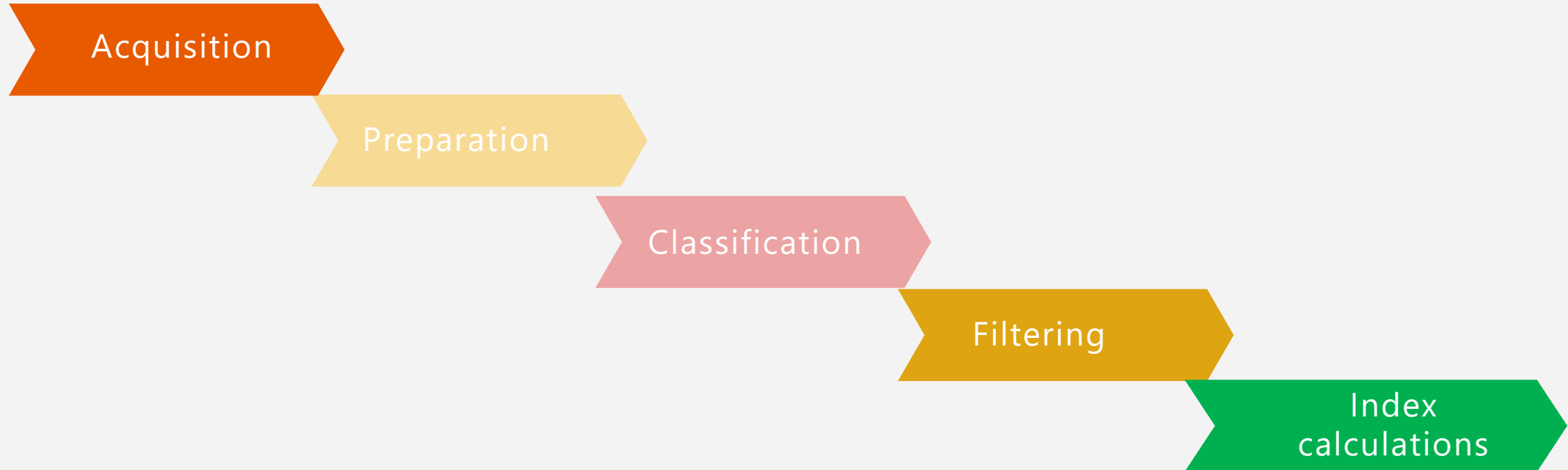


Production

Keep track of the procedures and quality indicators to make sure everything is ok. Keep track of procedures to check if they are working ok or need to be update. Make sure procedures are running properly and keep fallback plan up to date



Core procedures in an ADS pipeline



Core procedures

Acquisition

- Negotiations for data access
- Build tools for data transfer or extraction
- Track the quality of the transfer/extraction processes



Core procedures in an ADS pipeline



Acquisition

Preparation

- Data standardization
- Data aggregation
- Strategy and tools to identify and track products along time



Core procedures in an ADS pipeline



Acquisition

Preparation

Classification

- Inspect and identify internal classification structures of the retailer that can help to classify data into the CPI structure
- Identify proper classification method
- Develop and evaluate the robustness of the classification tools over time



Core procedures in an ADS pipeline



Acquisition

Preparation

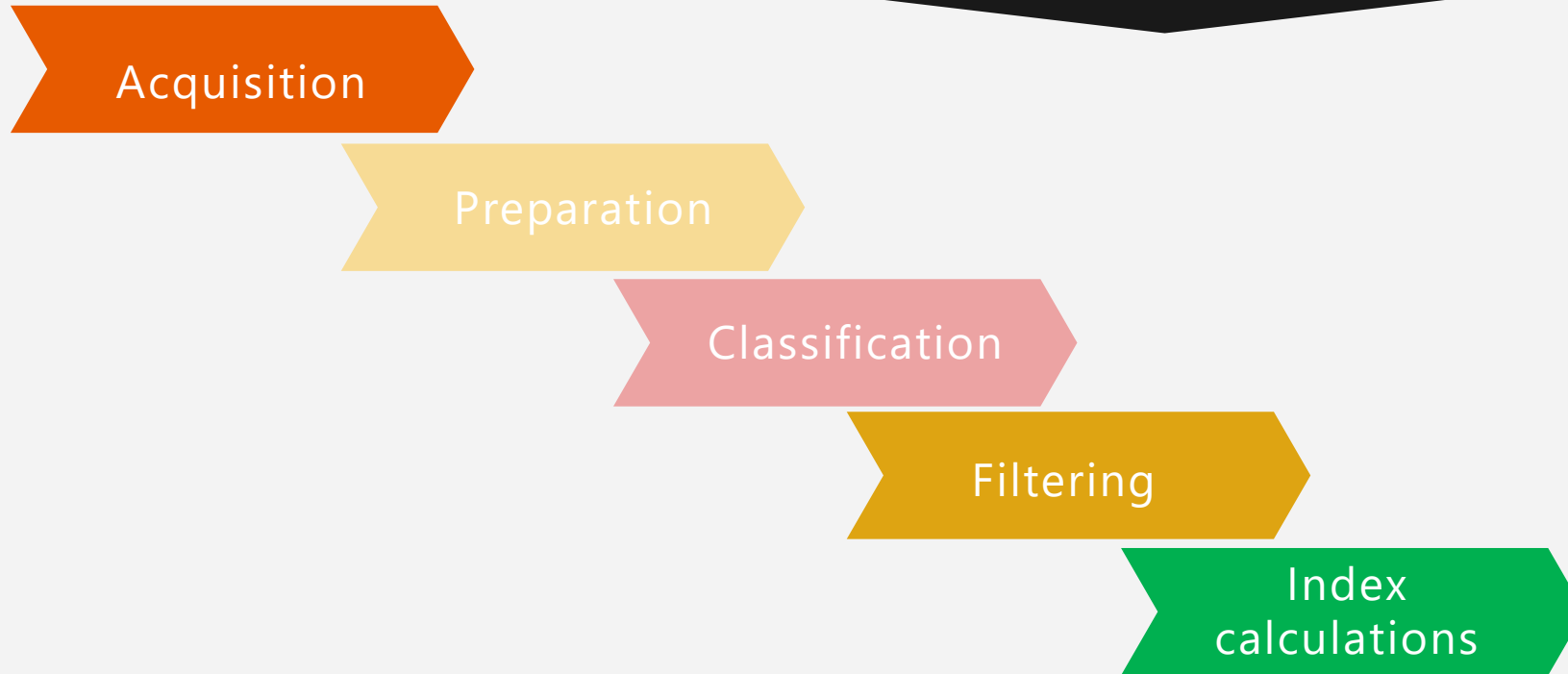
Classification

Filtering

- Define and test appropriate tools to treat outliers in the data sets.
- Track appropriateness of filters criteria along time
- Define if imputation methods should be used to treat flagged outliers.



Core procedures in an ADS pipeline



- Test and develop tools for more robust index formulations
- Define the most appropriate index formulation
- Track appropriateness of the index along time



Integration of ADS

Integration requirements



New strata

Refinement of CPI structure introducing strata by kind of data source;



New elementary indices

Incorporation of new elementary indices formulations



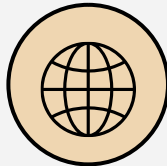
IT system changes

Changes in the CPI systems for the new calculations required



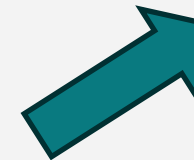
Staff routine

Changes in the CPI analysis routines.



Dissemination

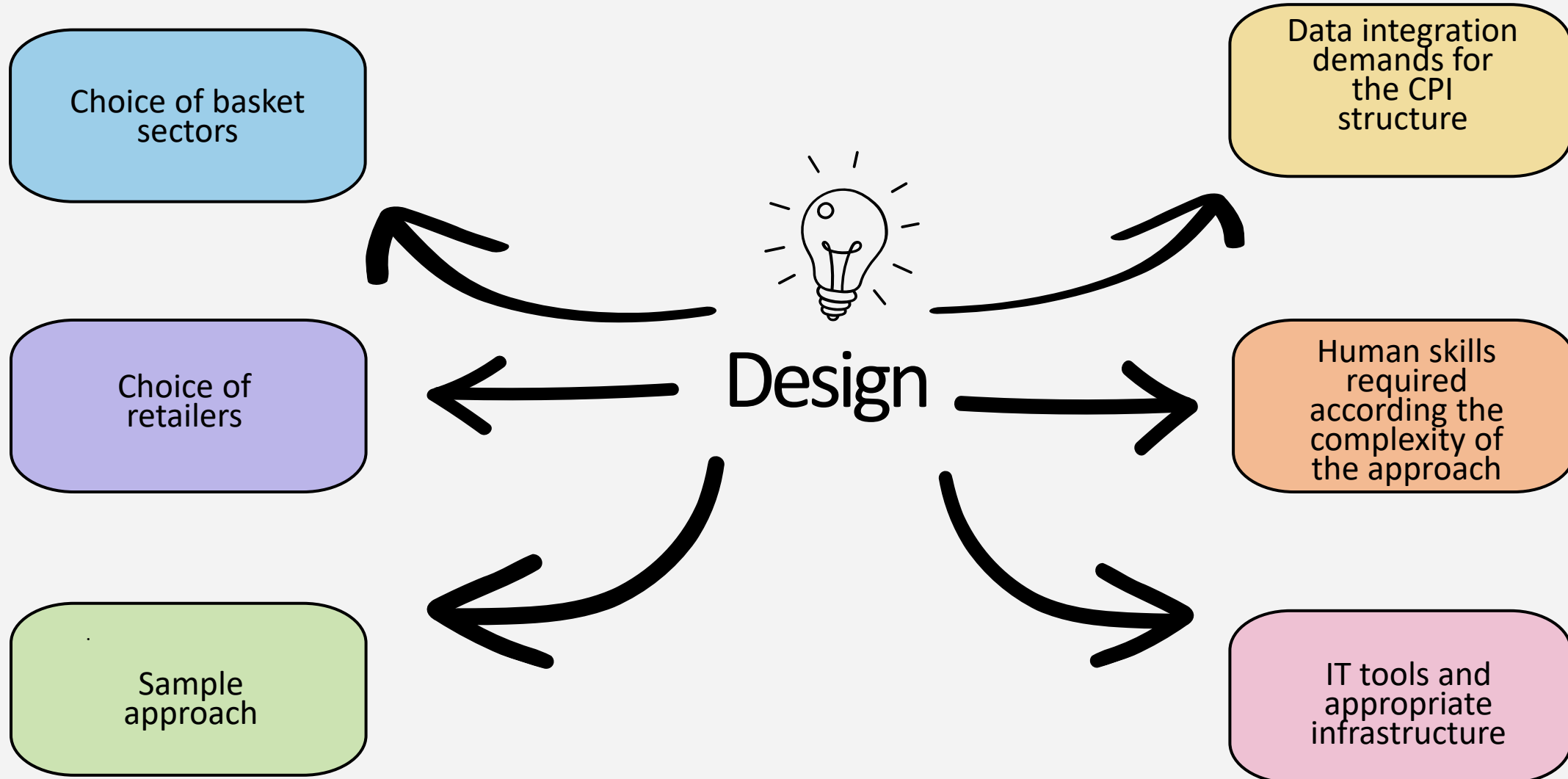
Evaluation of impact of changes when linking old and new series and Communication of changes to users



**Significant
extra
funding is
required**



“Before” start



“How” to start?

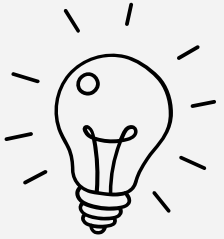
- Address problems with a reduced degree of complexity first;
- As maturity and expertise is developed, increase the complexity.



- Consider a reduced number of sectors and retailers first.
- Increase complexity as problems with the initial ones are solved and expertise gained.



How to start?



Points to
consider

Choice of basket sectors

Source availability

Representativity of
the sector;

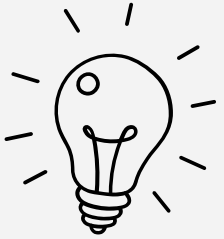
Value added

Market coverage;

Methodological
issues



How to start?



Points to consider

Choice of basket sectors

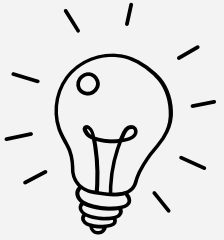
- Source availability
- Representativity of the sector;
- Value added
- Market coverage;
- Methodological issues

Choice of retailers

- Representativity in CPI sample
- Scope;
- Value added
- Market coverage



How to start?



Points to consider

Choice of basket sectors

- Source availability
- Representativity of the sector;
- Value added
- Market coverage;
- Methodological issues

Choice of retailers

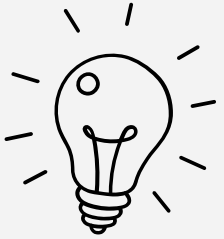
- Representativity in CPI sample and market coverage
- Scope;
- Value added

Sample approach

- Sample or universe of products?
- IT demands
- Availability of quantities



How to start?



Points to consider

Data integration demands

Weights availability

Degree of changes necessary

Impacts in the CPI

Derivation of new quality indicators

Human skills

Computer skills

Skills in prices indices

IT demands

System development

Construction of infrastructure

Use of external services



Conclusions

1. We have gone through and compared the most important features of different ADS used for CPI compilation
2. The different phases and core procedures of an ADS project were introduced.
3. Important considerations on data integration were discussed.
4. Inspection of important points to consider before starting and for the design of an ADS project were highlighted.



Thank you for the attention

